Table 1: Representation matrices

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A_1	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{010}:\begin{bmatrix}1\end{bmatrix}$	$2_{100}:\begin{bmatrix}1\end{bmatrix}$	$3^+_{111}:\begin{bmatrix}1\end{bmatrix}$	$3^+_{-11-1}:\begin{bmatrix}1\end{bmatrix}$
	$3^+_{1-1-1}:\begin{bmatrix}1\end{bmatrix}$	$3^+_{-1-11}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{111}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{1-1-1}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{-1-11}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{-11-1}:\begin{bmatrix}1\end{bmatrix}$
	$2_{110}:\begin{bmatrix}1\end{bmatrix}$	$2_{1-10}:\begin{bmatrix}1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^-_{100}:\begin{bmatrix}1\end{bmatrix}$	$2_{011}:\begin{bmatrix}1\end{bmatrix}$
	$2_{01-1}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{100}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{010}:\begin{bmatrix}1\end{bmatrix}$	$2_{101}:\begin{bmatrix}1\end{bmatrix}$	$4^-_{010}:\begin{bmatrix}1\end{bmatrix}$	$2_{-101}:\begin{bmatrix}1\end{bmatrix}$
A_2	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{010}:\begin{bmatrix}1\end{bmatrix}$	$2_{100}:\begin{bmatrix}1\end{bmatrix}$	$3^+_{111}:\begin{bmatrix}1\end{bmatrix}$	$3^+_{-11-1}:\begin{bmatrix}1\end{bmatrix}$
	$3^+_{1-1-1}:\begin{bmatrix}1\end{bmatrix}$	$3^+_{-1-11}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{111}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{1-1-1}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{-1-11}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{-11-1}:\begin{bmatrix}1\end{bmatrix}$
	$2_{110}:\begin{bmatrix}-1\end{bmatrix}$	$2_{1-10}:\begin{bmatrix}-1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$4^-_{100}:\begin{bmatrix}-1\end{bmatrix}$	$2_{011}:\begin{bmatrix}-1\end{bmatrix}$
	$2_{01-1}:\begin{bmatrix}-1\end{bmatrix}$	$4^+_{100}:\begin{bmatrix}-1\end{bmatrix}$	$4^+_{010}:\begin{bmatrix}-1\end{bmatrix}$	$2_{101}:\begin{bmatrix}-1\end{bmatrix}$	$4^-_{010}:\begin{bmatrix}-1\end{bmatrix}$	$2_{-101}:\begin{bmatrix}-1\end{bmatrix}$
E	$1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$2_{010}:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$2_{100}:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$3^+_{111}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$3^+_{-11-1}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$
	$3^+_{1-1-1}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$3^+_{-1-11}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$3^-_{111}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$3^-_{1-1-1}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$3^-_{-1-11}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$3^-_{-11-1}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$
	$2_{110}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$2_{1-10}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$4^-_{100}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$2_{011}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$
	$2_{01-1}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$4^+_{100}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$4^+_{010}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$2_{101}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$4^-_{010}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$2_{-101}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$
T_1	$1:\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1&0&0\\0&-1&0\\0&0&1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-1&0&0\\0&1&0\\0&0&-1\end{bmatrix}$	$2_{100}:\begin{bmatrix}1&0&0\\0&-1&0\\0&0&-1\end{bmatrix}$	$3^+_{111}:\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$	$3^+_{-11-1}:\begin{bmatrix}0&0&1\\-1&0&0\\0&-1&0\end{bmatrix}$
	$3^+_{1-1-1}:\begin{bmatrix}0&0&-1\\-1&0&0\\0&1&0\end{bmatrix}$	$3^+_{-1-11}:\begin{bmatrix}0&0&-1\\1&0&0\\0&-1&0\end{bmatrix}$	$3^-_{111}:\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}$	$3^-_{1-1-1}:\begin{bmatrix}0&-1&0\\0&0&1\\-1&0&0\end{bmatrix}$	$3^-_{-1-11}:\begin{bmatrix}0&1&0\\0&0&-1\\-1&0&0\end{bmatrix}$	$3^-_{-11-1}:\begin{bmatrix}0&-1&0\\0&0&-1\\1&0&0\end{bmatrix}$
	$2_{110}:\begin{bmatrix}0&1&0\\1&0&0\\0&0&-1\end{bmatrix}$	$2_{1-10}:\begin{bmatrix}0&-1&0\\-1&0&0\\0&0&-1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}0&1&0\\-1&0&0\\0&0&1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}0&-1&0\\1&0&0\\0&0&1\end{bmatrix}$	$4^-_{100}:\begin{bmatrix}1&0&0\\0&0&1\\0&-1&0\end{bmatrix}$	$2_{011}:\begin{bmatrix}-1&0&0\\0&0&1\\0&1&0\end{bmatrix}$
	$2_{01-1}:\begin{bmatrix}-1&0&0\\0&0&-1\\0&-1&0\end{bmatrix}$	$4^+_{100}:\begin{bmatrix}1&0&0\\0&0&-1\\0&1&0\end{bmatrix}$	$4^+_{010}:\begin{bmatrix}0&0&1\\0&1&0\\-1&0&0\end{bmatrix}$	$2_{101}:\begin{bmatrix}0&0&1\\0&-1&0\\1&0&0\end{bmatrix}$	$4^-_{010}:\begin{bmatrix}0&0&-1\\0&1&0\\1&0&0\end{bmatrix}$	$2_{-101}:\begin{bmatrix}0&0&-1\\0&-1&0\\-1&0&0\end{bmatrix}$
T_2	$1:\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1&0&0\\0&-1&0\\0&0&1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-1&0&0\\0&1&0\\0&0&-1\end{bmatrix}$	$2_{100}:\begin{bmatrix}1&0&0\\0&-1&0\\0&0&-1\end{bmatrix}$	$3^+_{111}:\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$	$3^+_{-11-1}:\begin{bmatrix}0&0&1\\-1&0&0\\0&-1&0\end{bmatrix}$
	$3^+_{1-1-1}:\begin{bmatrix}0&0&-1\\-1&0&0\\0&1&0\end{bmatrix}$	$3^+_{-1-11}:\begin{bmatrix}0&0&-1\\1&0&0\\0&-1&0\end{bmatrix}$	$3^-_{111}:\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}$	$3^-_{1-1-1}:\begin{bmatrix}0&-1&0\\0&0&1\\-1&0&0\end{bmatrix}$	$3^-_{-1-11}:\begin{bmatrix}0&1&0\\0&0&-1\\-1&0&0\end{bmatrix}$	$3^-_{-11-1}:\begin{bmatrix}0&-1&0\\0&0&-1\\1&0&0\end{bmatrix}$

continued ...

Table 1

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$2_{110}: \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$2_{1-10}: \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$4^-_{001}: \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	$4^+_{001}: \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	$4^-_{100}: \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$	$2_{011}: \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$
$2_{01-1}: \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$	$4^+_{100}: \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$	$4^+_{010}: \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	$2_{101}: \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$	$4^-_{010}: \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$	$2_{-101}: \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$